

ANDRÉS CASTILLO

RESEARCH SCIENTIST — FLUID MECHANICS, TWO-PHASE FLOW, PARAMETRIC INSTABILITIES

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RESEARCH PROFILE

Research scientist in fluid mechanics with 12+ years of experience in hydrodynamic instabilities, interfacial dynamics, and turbulent mixing. My recent work centres on fluids under periodic external forcing — Faraday waves and vibration-induced “frozen waves” — and on how that forcing shapes an interface and drives the mixing beneath it. I combine high-performance simulation, laboratory experiments, and reduced models, working closely with experimentalists and national-lab partners (CNRS, CEA, CNES).

SELECTED PUBLICATIONS

PARAMETRIC INSTABILITIES · TWO-PHASE FLOW · MIXING

- Mixing induced by Faraday surface waves — *pre-print*
Castillo-Castellanos, B.-J. Gréa, A. Briard, L. Gostiaux (2025)
- Frozen waves in the inertial regime — *Journal of Fluid Mechanics*
B.-J. Gréa, Castillo-Castellanos, A. Briard et al. (2025)

VORTEX DYNAMICS

- Simplified model for helical vortex dynamics in the wake of an asymmetric rotor — *Flow*
A. Abraham, Castillo-Castellanos, T. Leweke (2023)
- Closely spaced corotating helical vortices: long-wave instability — *Journal of Fluid Mechanics*
Castillo-Castellanos, S. Le Dizès (2022)
- Closely spaced corotating helical vortices: general solutions — *Physical Review Fluids*
Castillo-Castellanos, S. Le Dizès, E. Durán Venegas (2021)

TURBULENT CONVECTION

- Cessation and reversals of large-scale structures in square Rayleigh–Bénard cells — *Journal of Fluid Mechanics*
Castillo-Castellanos, A. Sergent, B. Podvin, M. Rossi (2019)
 - Reversal cycle in square Rayleigh–Bénard cells in the turbulent regime — *Journal of Fluid Mechanics*
Castillo-Castellanos, A. Sergent, M. Rossi (2016)
 - Turbulent convection in Rayleigh–Bénard cells with modified boundary conditions — *PhD thesis*
Castillo-Castellanos (2017), Université Pierre-et-Marie-Curie
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RESEARCH POSITIONS

Researcher — Centre Borelli

Since 2022 · Paris-Saclay, France
CNRS/École Normale Supérieure Paris-Saclay (UMR 9010). Work carried out within LRC MESO, a joint laboratory with CEA.

- Currently studying how surface tension and viscosity contrast alter the growth and mixing of the Rayleigh–Taylor instability, building a database of high-fidelity simulations to drive machine-learning models at CEA.
- Studied how two immiscible fluids reach equilibrium under intense horizontal vibration (“frozen waves”). Slated to join a Zero-G campaign led by S. Fauve and M. Berhanu to study frozen waves in microgravity (71st CNES parabolic flight campaign, VP194, October 2026).

- Showed how Faraday surface waves inject energy into the stratified layer beneath them and mix it through turbulent entrainment, building simple models from joint experiments and simulations.

Key collaborators: B.-J. Gréa^{CEA}, A. Briard^{CEA}, L. Gostiaux^{CNRS}

Researcher — Institut de Recherche sur les Phénomènes Hors Équilibre (IRPHE)

2019–2022 · Marseille, France

Institute for Research on Non-Equilibrium Phenomena, CNRS/Aix-Marseille Université (UMR 7342).

- Modelled how the wake behind a rotor breaks down, showing that vortex instabilities can be deliberately triggered to accelerate wake recovery by introducing a small asymmetry or modulating the rotation rate.
- Showed how closely spaced pairs of helical vortices deform and become unstable (long-wave instabilities), generalised to arbitrary geometries.

Key collaborators: S. Le Dizès^{CNRS}, T. Leweke^{CNRS}

Researcher — Laboratoire d'informatique pour la mécanique et les sciences de l'ingénieur (LIMSI, now LISN)

2017–2018 · Orsay, France

CNRS (UPR 3251), Orsay, France.

- Built reduced-order models that capture the essential dynamics of flow reversals in turbulent convection (POD, CROM).

Key collaborators: A. Sergent^{Sorbonne Université}, B. Podvin^{CNRS}

PhD student — Institut Jean Le Rond d'Alembert (IJLRDA)

2013–2017 · Paris, France

CNRS/Sorbonne Université (UMR 7190), Paris, France.

- Characterised rare, sudden reversals of the large-scale circulation in turbulent convection, using direct numerical simulation (DNS).
- Showed how lateral confinement and boundary conditions change the flow structure and heat transfer in turbulent convection.

Key collaborators: M. Rossi^{CNRS}, A. Sergent^{Sorbonne Université}

EDUCATION

PhD in Fluid Mechanics — Université Pierre-et-Marie-Curie (now Sorbonne Université)

2017 · Paris

MSc in Energetics & Environment — Arts et Métiers (ENSAM)

2013 · Paris

BSc in Mechanical Engineering — Universidad Industrial de Santander

2008 · Bucaramanga, Colombia

TECHNICAL SKILLS & LANGUAGES

Numerical methods — FEM, FVM, DNS, reduced-order models, linear stability analysis

CFD — Basilisk, Ansys CFX, Fluent

Programming — Python, C, Fortran, MATLAB, Bash, Git

HPC & debugging — GDB, DDT, Scalasca, HDF5

Visualisation — ParaView, VisIt, Matplotlib, Mayavi

Other — OpenCV, Ansys APDL, Autodesk Inventor, AutoCAD

Spanish (native); English, French (full professional proficiency).